

Homework Assignment #2 on the topics “Game theory”

This homework should be completed in groups of 2-4 people. To complete the homework, you need to analyse a real-life strategic decision-making problem using concepts from game theory and empirical data.

To complete the homework, you need to:

1. Select a real-life problem that involves strategic interaction between two or more players. The problem should be relevant to everyday, professional, economic, social, sports-related or data-driven situations and supported by real empirical data. During the lecture, we discussed several examples of strategic interactions, including prisoner's dilemma, the free rider dilemma, pricing decisions in economics, negotiations and conflicts, fraud detection problems, A/B testing, penalty kicks in football, and strategic behaviour in board or card games. You may take inspiration from these examples, but the analysis should not be purely hypothetical. You should find, collect or estimate real data and use it to estimate strategies, payoffs, probabilities frequencies or other quantities required for the game-theoretic model.
 - a) Provide a detailed description of the selected problem, including the context, players, strategies, possible outcomes and data sources.
 - b) Explain why the problem is important or interesting from a game-theoretic perspective.
 - c) Clearly state whether the problem is modelled as a simultaneous/static game or as a dynamic/sequential game.
 - d) The data should contain a sufficient number of observations to support the analysis.
2. Based on the selected problem, formulate the situation as a game. Define the players, strategies, outcomes and payoffs. Explain how the payoffs are calculated or estimated from the data. For a static game, construct a payoff matrix. For a dynamic/sequential game, construct a game tree. Explain all assumptions used in the model.
3. Solve the game using an appropriate method. Depending on the type of game selected, apply the appropriate game-theoretic solution method. For instance, for static games, find the Nash equilibrium in pure strategies and, if relevant, in mixed strategies. Compare the theoretical prediction with the empirical behaviour observed in the data.
4. Analyse and interpret the obtained results. Does the theoretical solution coincide with the empirical behaviour? Which player behaves closer to the theoretical prediction? Are there deviations from the theoretical solution? What practical conclusions can be drawn? What are the limitations of the model and data?

Based on your assignment, please prepare a report in Word/PDF format that includes a complete solution with detailed explanations and the results for all parts of the assignment, including the data, payoff calculations, and formulas used. You can use Excel, Jupyter Notebooks, Python, R, or other software tools to solve the task. In that case, ensure that you attach the source files, datasets and calculation/code files. Do not forget to include the full names of all group members.

The completed report should be uploaded to the SmartLMS system, section “Homework Assignment #2”. **The deadline is 23:59 on May 27, 2026.**